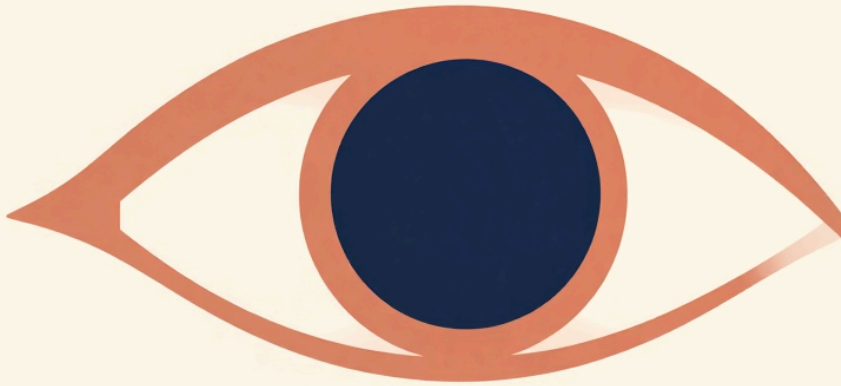




PAPER EXPLAINED

Detect blinks faster with linear transformers

BlinkLinMult: Transformer-Based Eye Blink Detection

https://adamfodor.com/pdf/2023_Fodor_Adam_MDPI_BlinkLinMult.pdf

The Blink Detection Challenge

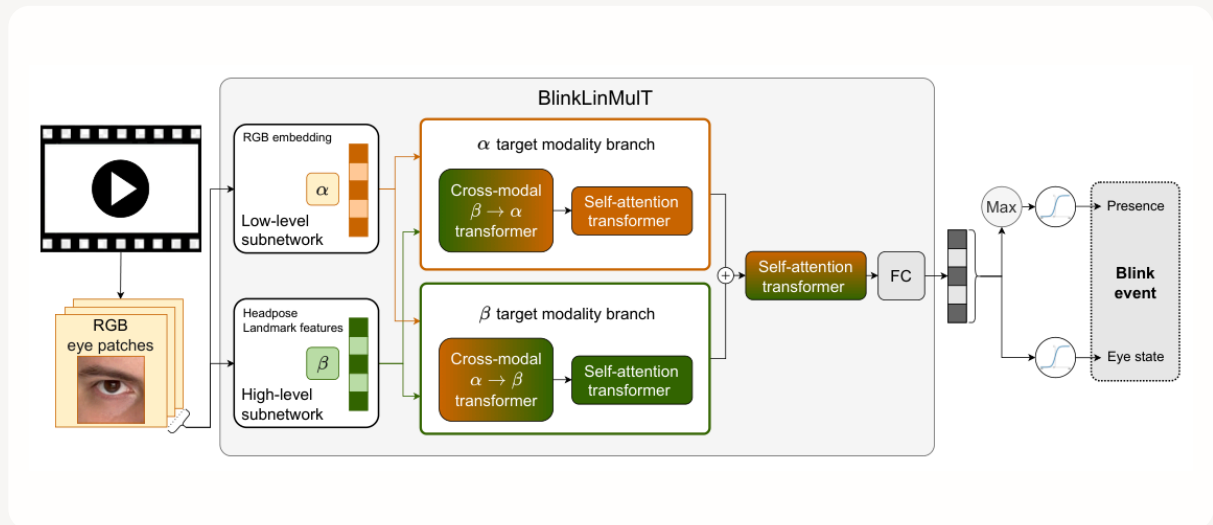
- Detecting eye blinks is challenging but reveals deep fakes and driver drowsiness.
- Existing methods struggle with short blinks due to frame-wise classification.

Introducing BlinkLinMult

- The authors propose a fast transformer-based framework for eye blink detection.
- It combines low- and high-level feature sequences including head pose angles.

Linear Attention Architecture

- BlinkLinMulT uses a modified multi-modal transformer with linear attention to process RGB texture and landmarks.
- The model replaces quadratic attention modules with linear versions for easier training while maintaining similar performance.
- A backbone CNN extracts features from eye patches, which a linear transformer uses to estimate blinks for each eye independently.



BlinkLinMulT: a multimodal transformer architecture for blink presence detection and frame-wise eye state recognition.

Superior Cross-Dataset Results

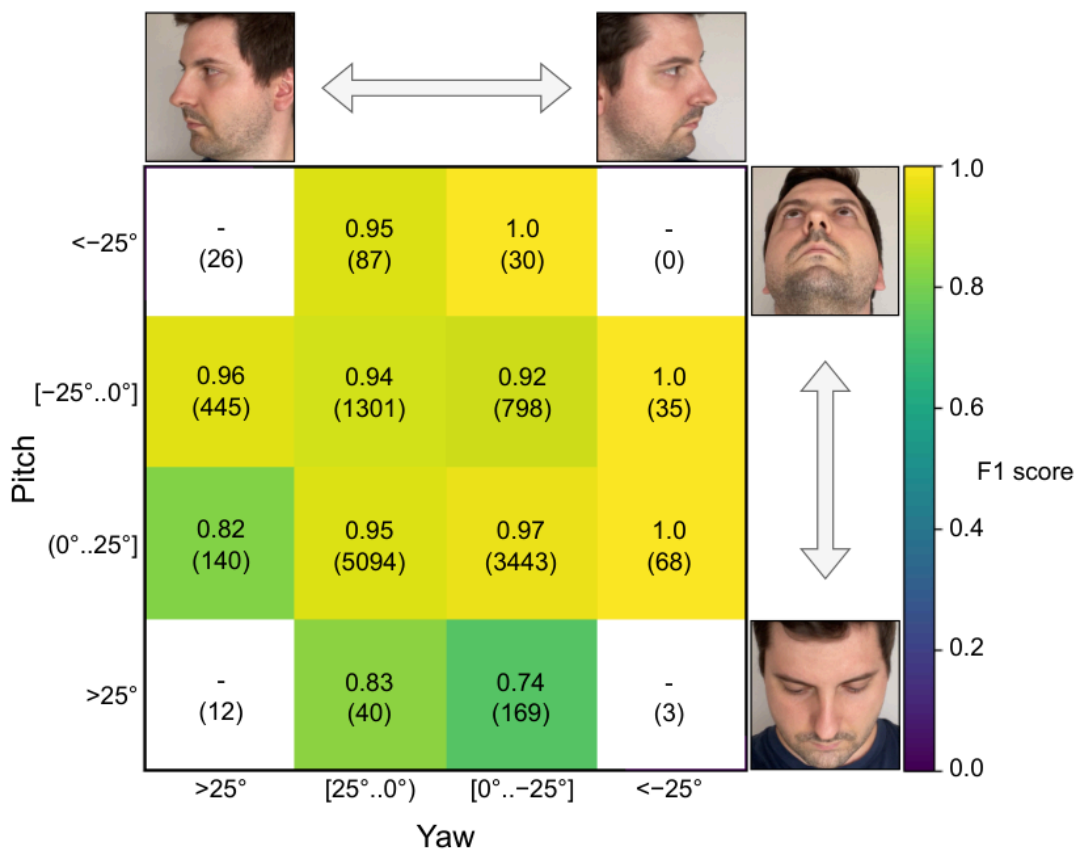
- BlinkLinMulT matches or beats state-of-the-art methods across various datasets.
- It achieves a perfect 1.0 F1 score for blink detection on the TalkingFace dataset.

Robustness via Union Training

- Union training improved F1 scores on CEW, ZJU, and RT-BENEimg benchmarks.
- Eye state recognition and blink detection outperformed single dataset baselines.

Handling Extreme Head Poses

- Model performance drops at higher yaw angles.
- Performance slightly decreases at extreme orientations.



Head pose angle dependence of BlinkLinMulT in the case of blink presence detection task.

Unified Training for Generalization

- The BlinkLinMulT model is trained once on the union of all datasets.
- This approach improves robustness and results across all individual unseen datasets.